Christoph Johannes Jabs

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Education and Training

Academic Education

08/2020–06/2022 Master of Science in Computer Science, University of Helsinki (FI),
Final grade: 5 (Grading scale from 5, 'excellent', to 0, 'fail')
Thesis grade: 5

04/2020–08/2020 Select courses in Computer Science, Eberhard Karls University, Tübingen (DE)

03/2016–02/2020 Bachelor of Engineering in Mechatronics, Reutlingen University (DE),
Final grade: 1.0 (Grading scale from 1.0, 'very good', to 5.0, 'fail')
Thesis grade: 1.0

09/2008–06/2015 High School Diploma Karl-von-Frisch Gymnasium, Dusslingen (Germany)
Final grade: 1.1 (Grading scale from 1.0, 'very good', to 6.0, 'fail')

Professional Training

09/2015–07/2017 Apprenticeship as a Mechatronic Technician, Robert Bosch GmbH, Reutlingen (DE), Combined with Bachelor's studies

Professional Experience

09/2022–Current Doctoral researcher at the Universität Helsinki (FI)
Supervised by Matti Järvisalo in the Constraint Reasoning and Optimization research group
- (Multi-objective) declarative optimization
- Fair data representation

05/2021–08/2022 Research Assistant at the University of Helsinki (FI),
Constraint Reasoning and Optimization research group
- Investigating exact bi-objective optimization
- Exploring applications of incremental MaxSAT solving

10/2019–02/2020 Bachelor's Thesis at Robert Bosch GmbH, Reutlingen (DE)
'A Recurrent Neural Net Approach to Activity Recognition'

11/2017–06/2020 Student Jobs and Internships at Robert Bosch GmbH, Reutlingen (DE)
- Planned and executed data collection for mobile theft detection for eBikes
- Planned and implemented automation for acoustics measurements of eBikes
- Integrated an open-source Bluetooth stack into a Python testing framework
- Implemented software for automatic measurement report generation
- Implemented software for validating automotive electronic control units

Publications

2022a. 'A Maximum Satisfiability Based Approach to Bi-Objective Boolean Optimization'. M. Sc. thesis. University of Helsinki. URL: `http://urn.fi/URN:NBN:fi:hulib-202206132323`.

2022b (with Jeremias Berg, Andreas Niskanen and Matti Järvisalo). 'MaxSAT-Based Bi-Objective Boolean Optimization'. In: *SAT*. Vol. 236. LIPIcs. Schloss Dagstuhl — Leibniz-Zentrum für Informatik, 12:1–12:23. DOI: `10.4230/LIPIcs.SAT.2022.12`. [Single blind peer review].

2020. 'A Recurrent Neural Net Approach to Activity Recognition'. B. Eng. thesis. Reutlingen University.

Presentations

08/2022 'MaxSAT-Based Bi-Objective Boolean Optimization' at *SAT 2022* in Haifa, Israel

Professional Skills

Languages

German — Mother tongue

English — Effective operational proficiency, TOEFL iBT 114

French — Intermediate, DELF B1

Computer Skills

Operating Systems — Linux, Microsoft Windows

Word Processing — LaTeX, Markup, Microsoft Word, LibreOffice Writer

Programming Languages — C/C++, Python, Bash, Matlab

Training and Workshops

08/2022	13th 'Pragmatics of SAT' workshop at *FLoC 2022* in Haifa, Israel
06/2021	'Introduction to Scientific Computing' by *Aalto scientific computing*, online

Conference Visits

08/2022	*Federated Logic Conference (FLoC)* in Haifa, Israel, including the 25th *International Conference on Theory and Applications of Satisfiability Testing (SAT)*
07/2022	32nd *European Conference on Operational Research (EURO)* in Espoo, Finland

Voluntary Work

08/2022	Volunteering at *Federated Logic Conference 2022* in Haifa, Israel
2013–2020	Volunteering youth work in a YMCA and the Lutheran Church in Germany